

My Project

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Chapter 1

Class Index

1.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

patient	Stores all information about a patient	5
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Chapter 2

File Index

2.1 File List

Here is a list of all documented files with brief descriptions:

dentist.c	Dentist Clinic Simulation - Operating System Project	9
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Chapter 3

Class Documentation

3.1 patient Struct Reference

Stores all information about a patient.

Public Attributes

- int [id](#)
- char [name](#) [64]
- int [appointment](#)
- int [willArrive](#)
- int [arrivalOrder](#)
- int [waiting](#)
- int [treated](#)
- int [noShow](#)
- int [leftNoChair](#)
- int [leftDoctorFull](#)
- int [requestPending](#)
- sem_t [replySem](#)
- sem_t [dentistCallSem](#)
- sem_t [treatmentDoneSem](#)

3.1.1 Detailed Description

Stores all information about a patient.

This structure stores appointment information, waiting room information, treatment status and private semaphores.

3.1.2 Member Data Documentation

3.1.2.1 appointment

```
int patient::appointment
```

1 = confirmed, 0 = not confirmed

3.1.2.2 arrivalOrder

```
int patient::arrivalOrder
```

FIFO / FCFS arrival order

3.1.2.3 dentistCallSem

```
sem_t patient::dentistCallSem
```

Dentist call semaphore

3.1.2.4 id

```
int patient::id
```

Patient ID

3.1.2.5 leftDoctorFull

```
int patient::leftDoctorFull
```

1 = left because doctor was full

3.1.2.6 leftNoChair

```
int patient::leftNoChair
```

1 = left because room was full

3.1.2.7 name

```
char patient::name[64]
```

Patient name

3.1.2.8 noShow

```
int patient::noShow
```

1 = no-show

3.1.2.9 replySem

```
sem_t patient::replySem
```

Receptionist reply semaphore

3.1.2.10 requestPending

```
int patient::requestPending
```

1 = appointment request pending

3.1.2.11 treated

```
int patient::treated
```

1 = treated

3.1.2.12 treatmentDoneSem

```
sem_t patient::treatmentDoneSem
```

Treatment completion semaphore

3.1.2.13 waiting

```
int patient::waiting
```

1 = waiting, 0 = not waiting

3.1.2.14 willArrive

```
int patient::willArrive
```

1 = will arrive, 0 = no-show

The documentation for this struct was generated from the following file:

- [dentist.c](#)

Chapter 4

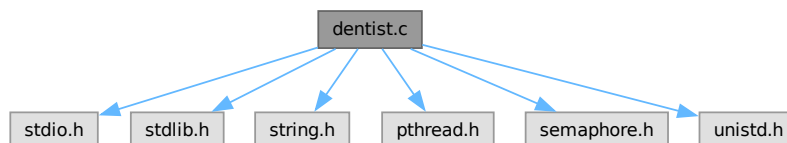
File Documentation

4.1 dentist.c File Reference

Dentist Clinic Simulation - Operating System Project.

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>
```

Include dependency graph for dentist.c:



Classes

- struct `patient`
Stores all information about a patient.

Functions

- void `initializePatient` (int i)
Initializes a patient structure.
- void `destroyPatient` (int i)
Destroys all semaphores of a patient.
- int `findNextPatient` ()
Finds the next patient using FIFO / FCFS.

- int `findNextRequest` ()
Finds the next appointment request.
- void * `receptionist` (void *arg)
Receptionist thread function.
- void * `dentist` (void *arg)
Dentist thread function.
- void * `patient` (void *arg)
Patient thread function.
- void `printPatientTable` ()
Displays the final patient table.
- void `printStatistics` ()
Displays final clinic statistics.
- void `showProjectInfo` ()
Displays the contents of project_info.txt.
- int `main` ()
Main entry point of the program.

Variables

- int `totalPatients`
- int `waitingRoomChairs`
- int `doctorCapacity`
- int `appointmentSlots`
- int `waitingPatients` = 0
- int `treatedPatients` = 0
- int `confirmedAppointments` = 0
- int `activeAppointments` = 0
- int `arrivalCounter` = 0
- int `clinicClosed` = 0
- struct `patient` * `patients`
Pointer to dynamically allocated patient array.
- pthread_t * `patientThreads`
Pointer to dynamically allocated patient thread array.
- pthread_mutex_t `mutex`
Mutex used to protect shared data.
- sem_t `requestReady`
Semaphore used to notify the receptionist when a patient requests an appointment.
- sem_t `patientReady`
Semaphore used to notify the dentist when a patient enters the waiting room.

4.1.1 Detailed Description

Dentist Clinic Simulation - Operating System Project.

Author

shanjeda alam

Version

1.0

Date

2026

This program simulates a dentist clinic using different Operating System concepts.

Main OS Concepts:

1. POSIX Threads
2. Mutex
3. Semaphores
4. FIFO / FCFS Scheduling
5. Critical Section
6. Synchronization
7. Structure
8. Pointer
9. Dynamic Memory Allocation

Main Features:

- Patient management
- Appointment management
- Waiting room management
- Doctor treatment management
- No-show patient handling
- FIFO / FCFS scheduling
- Thread synchronization
- Mutex protected critical sections
- Semaphore based communication

User Input:

- Number of patients
- Number of waiting room chairs
- Doctor capacity
- Number of appointment slots
- Patient names
- Patient arrival / no-show status

Special Rule: If a confirmed appointment patient becomes a no-show, the appointment slot is released and another waiting patient can receive treatment.

4.1.2 Function Documentation

4.1.2.1 dentist()

```
void * dentist (  
    void * arg)
```

Dentist thread function.

Dentist thread.

Selects patients using FIFO / FCFS and treats them.

Parameters

<i>arg</i>	Thread argument.
------------	------------------

Returns

NULL when the thread finishes.

The dentist selects waiting patients according to FIFO/FCFS. Each patient receives treatment for a simulated period.

Parameters

<i>arg</i>	Thread argument.
------------	------------------

Returns

NULL when dentist exits.

4.1.2.2 destroyPatient()

```
void destroyPatient (  
    int i)
```

Destroys all semaphores of a patient.

Destroys patient semaphores.

Parameters

<i>i</i>	Index of the patient.
----------	-----------------------

4.1.2.3 findNextPatient()

```
int findNextPatient ()
```

Finds the next patient using FIFO / FCFS.

Finds the next waiting patient using FIFO / FCFS.

Returns

Patient index if available, otherwise -1.

The patient with the smallest arrival order is selected.

Returns

Patient index or -1 if no patient is waiting.

4.1.2.4 findNextRequest()

```
int findNextRequest ()
```

Finds the next appointment request.

Finds the next patient requesting an appointment.

Returns

Patient index if request exists, otherwise -1.

Patient index or -1 if no request exists.

4.1.2.5 initializePatient()

```
void initializePatient (  
    int i)
```

Initializes a patient structure.

Initializes one patient.

Parameters

<i>i</i>	Index of the patient.
----------	-----------------------

All patient status values are set to their default values. Private semaphores are also initialized here.

Parameters

<i>i</i>	Index of the patient in the patient array.
----------	--

4.1.2.6 main()

```
int main ()
```

Main entry point of the program.

Takes clinic settings and patient information from the user, initializes threads, mutexes and semaphores, starts the simulation and finally cleans up all resources.

Returns

0 if the program finishes successfully.

4.1.2.7 patient()

```
void * patient (  
    void * arg)
```

Patient thread function.

Patient thread.

Handles appointment request, arrival, waiting room, no-show and treatment.

Parameters

<i>arg</i>	Pointer to a patient structure.
------------	---------------------------------

Returns

NULL when the thread finishes.

Handles:

- Appointment request
- Appointment confirmation
- No-show
- Waiting room entry
- FIFO order
- Treatment

Parameters

<i>arg</i>	Pointer to struct patient.
------------	----------------------------

Returns

NULL when patient thread finishes.

4.1.2.8 printPatientTable()

```
void printPatientTable ()
```

Displays the final patient table.

Displays final results of all patients.

4.1.2.9 receptionist()

```
void * receptionist (  
    void * arg)
```

Receptionist thread function.

Receptionist thread.

Handles appointment requests from patients.

Parameters

<i>arg</i>	Thread argument.
------------	------------------

Returns

NULL when the thread finishes.

The receptionist waits for appointment requests and assigns available appointment slots.

Parameters

<i>arg</i>	Thread argument.
------------	------------------

Returns

NULL when receptionist exits.

4.1.2.10 showProjectInfo()

```
void showProjectInfo ()
```

Displays the contents of project_info.txt.

Reads and displays project_info.txt.

This function connects the external text documentation file with the C program.

4.1.3 Variable Documentation

4.1.3.1 activeAppointments

```
int activeAppointments = 0
```

Currently active appointment slots.

4.1.3.2 appointmentSlots

```
int appointmentSlots
```

Number of available appointment slots.

4.1.3.3 arrivalCounter

```
int arrivalCounter = 0
```

Counter used for FIFO / FCFS order.

4.1.3.4 clinicClosed

```
int clinicClosed = 0
```

Indicates whether the clinic is closed.

4.1.3.5 confirmedAppointments

```
int confirmedAppointments = 0
```

Total number of confirmed appointments.

4.1.3.6 doctorCapacity

```
int doctorCapacity
```

Maximum number of patients the doctor can treat.

4.1.3.7 mutex

```
pthread_mutex_t mutex
```

Mutex used to protect shared data.

This mutex protects critical sections where multiple threads access shared variables.

4.1.3.8 totalPatients

```
int totalPatients
```

Total number of patients.

4.1.3.9 treatedPatients

```
int treatedPatients = 0
```

Number of patients already treated.

4.1.3.10 waitingPatients

```
int waitingPatients = 0
```

Current number of patients in waiting room.

4.1.3.11 waitingRoomChairs

```
int waitingRoomChairs
```

Number of chairs in the waiting room.

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